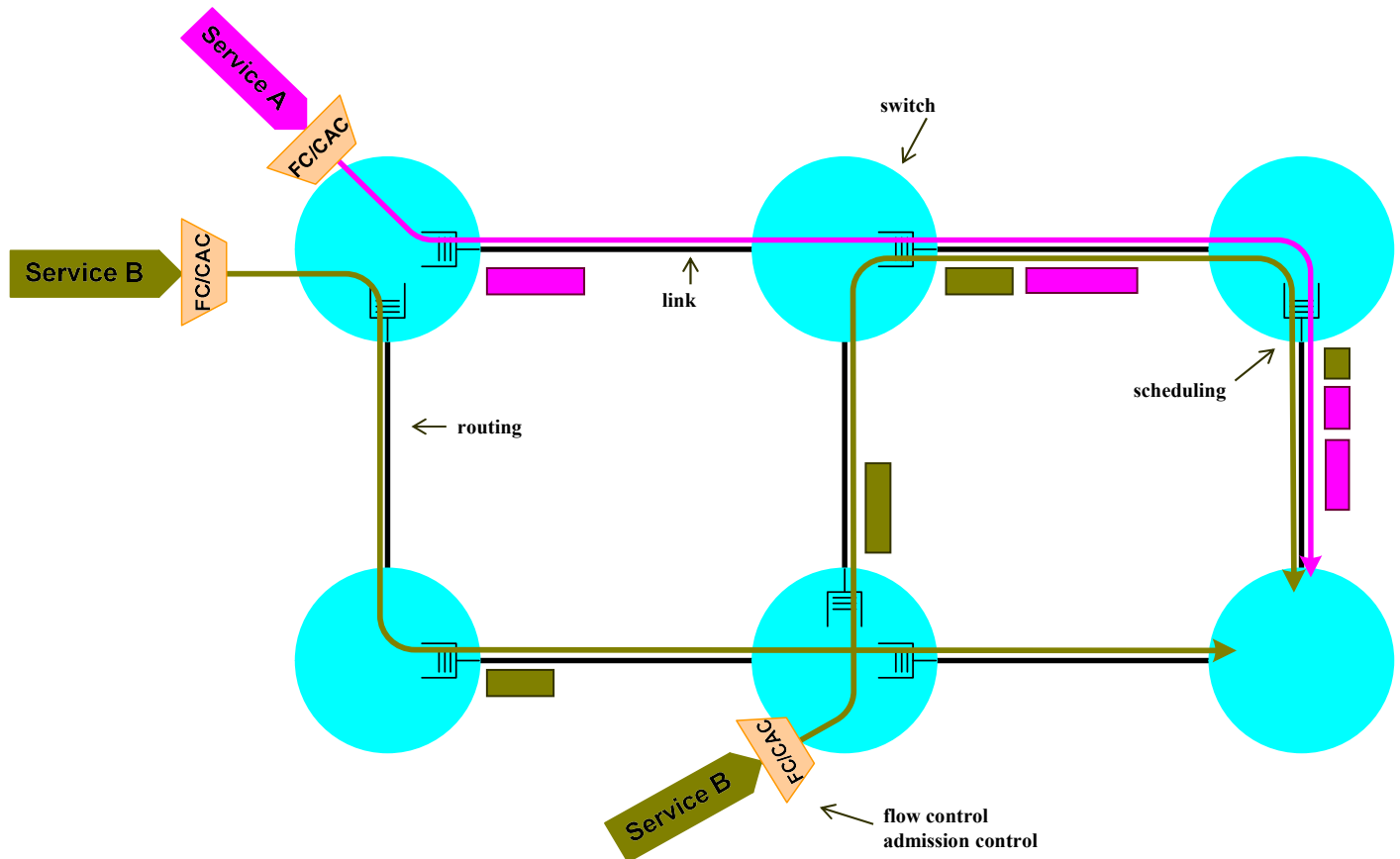


Packet-switched networks

Performance of packet-switched networks

- Depends on many factors:
 - The volume and stochastic characteristics of offered traffic
 - The way offered traffic is controlled at the network entrance – traffic shaping, congestion control, admission control
 - The way offered traffic is routed inside the network – routing
 - The QoS differentiation mechanisms – packet classification, scheduling and discard

Performance of packet-switched networks



Routing

- Function that determines the path followed by packets from origin to destination
- For a given packet flow (from some origin to some destination), routing can be:
 - Fixed – packets follow only one path
 - Bifurcated – packets are split among several possible paths
 - Dynamic – the path may vary according to state of the network
- Using longer paths may imply higher delays (but it also may not, due to different link capacities and to congestion state of some network zones)

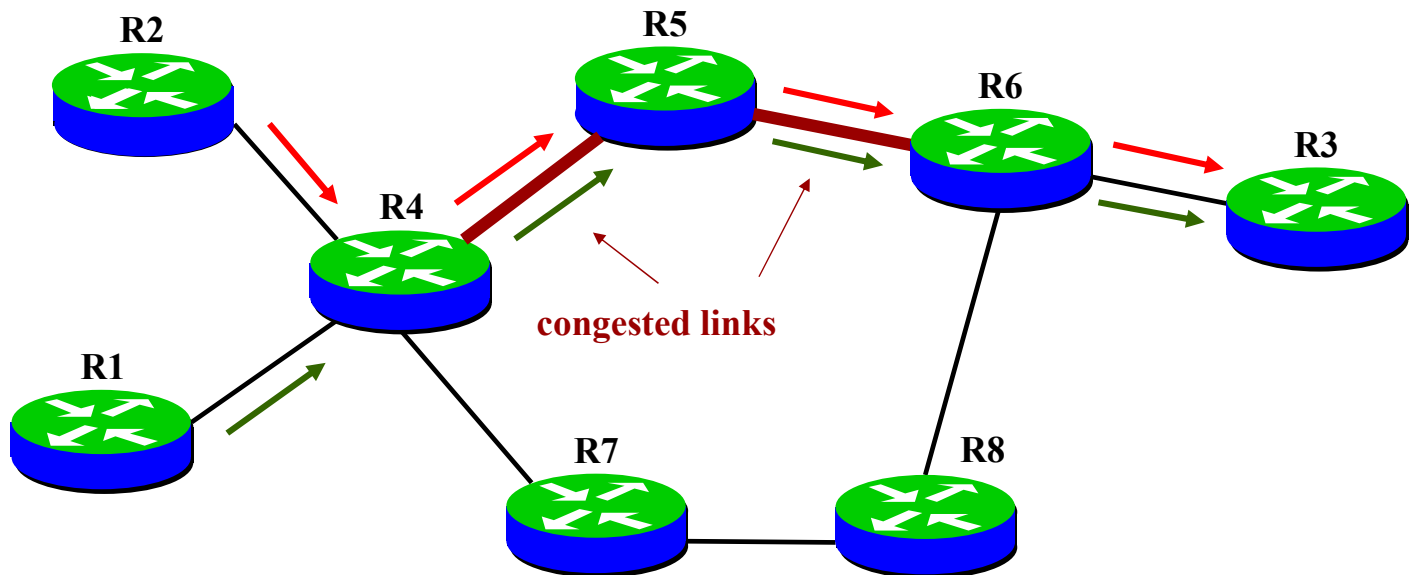
^
the

Internet routing

- Packets follow the **shortest path** to destination
- Several possibilities, according to specific routing protocol
 - Minimum hop routing (RIP)
 - User-defined weights (OSPF)
 - Equal-cost splitting (OSPF)
- But routes always tied to the destination (except if MPLS is used)

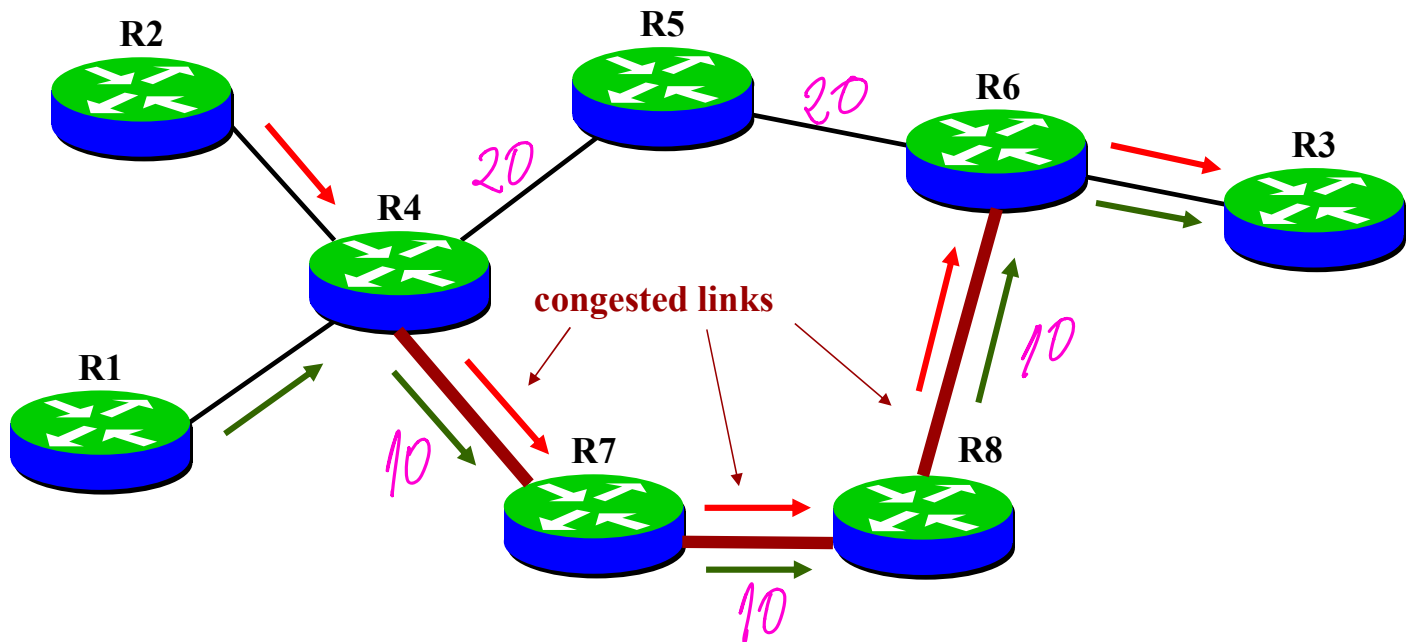
Minimum hop routing

- ⇒ Links have 150 Mb/s of capacity, except R6-R3
- ⇒ R6-R3 has 2 Gb/s of capacity
- ⇒ Flow R1-R3: 100 Mb/s
- ⇒ Flow R2-R3: 100 Mb/s



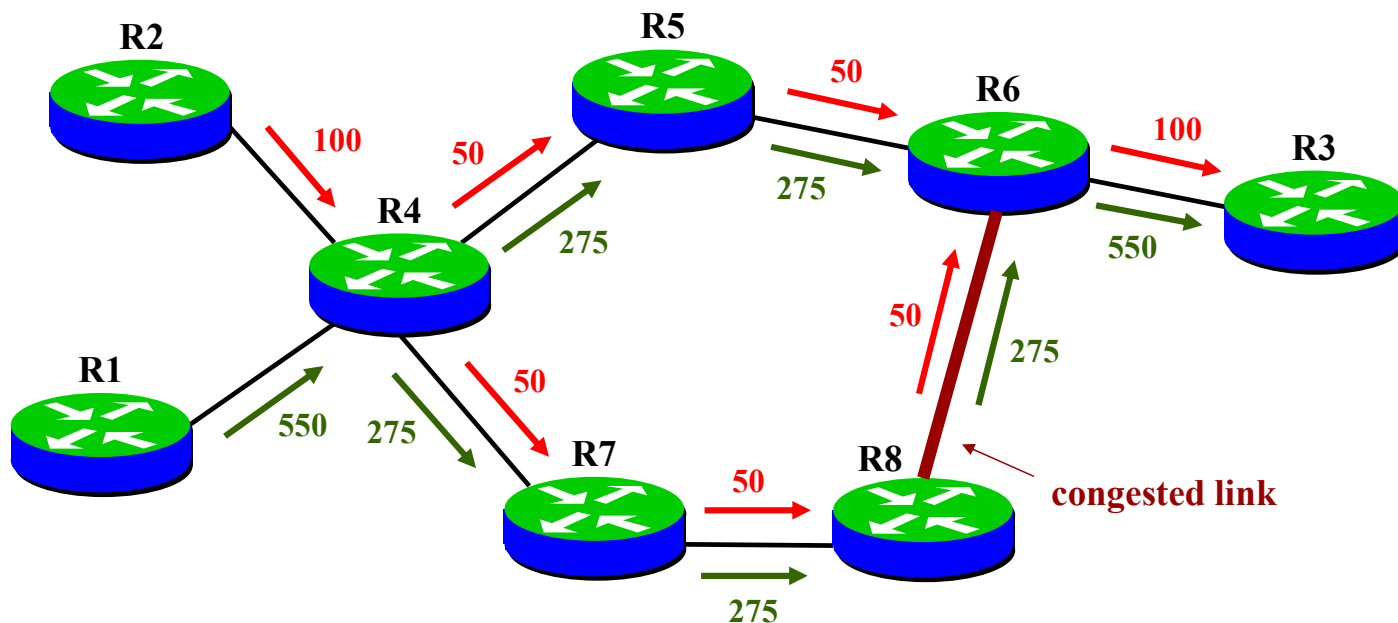
User-defined weights

- ⇒ Links have 150 Mb/s of capacity, except R6-R3
- ⇒ R6-R3 has 2 Gb/s of capacity
- ⇒ Flow R1-R3: 100 Mb/s
- ⇒ Flow R2-R3: 100 Mb/s



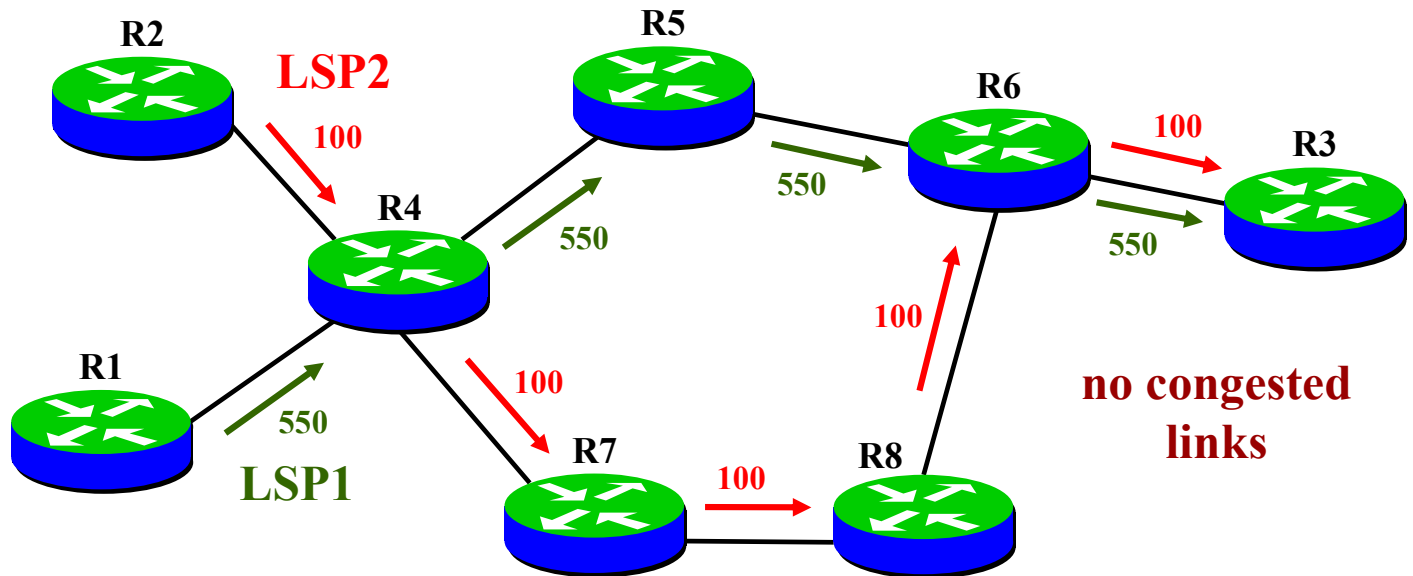
Equal-cost splitting

- ⇒ Links have 620 Mb/s of capacity, except R6-R3 and R8-R6
- ⇒ R6-R3 has 2 Gb/s; R8-R6 has 150 Mb/s
- ⇒ Flow R1-R3: 550 Mb/s
- ⇒ Flow R2-R3: 100 Mb/s
- ⇒ Administratively, cost of R4-R5-R6 = cost R4-R7-R8 *- R6*



Routing through LSPs (MPLS)

- ⇒ Links have 620 Mb/s of capacity, except R6-R3 and R8-R6
- ⇒ R6-R3 has 2 Gb/s; R8-R6 has 150 Mb/s
- ⇒ Flow R1-R3: 550 Mb/s
- ⇒ Flow R2-R3: 100 Mb/s
- ⇒ Each flow is transported in a different LSP

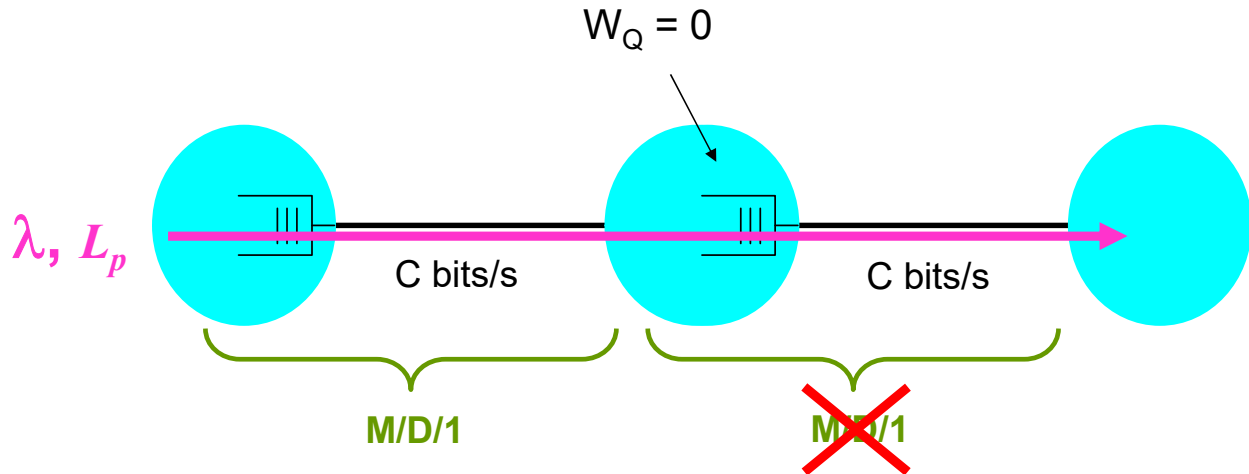


Routing is no longer tied to the destination address!

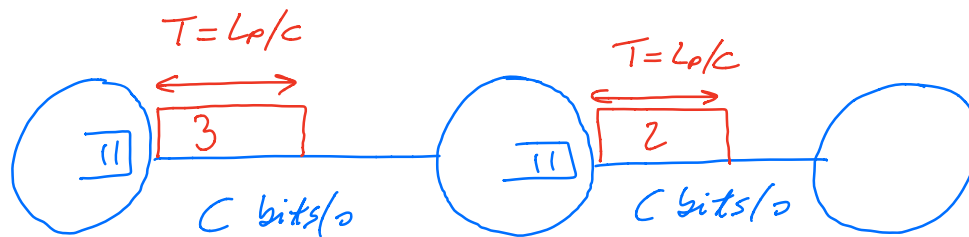
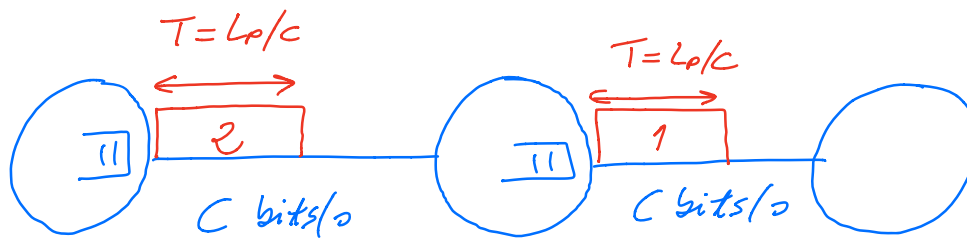
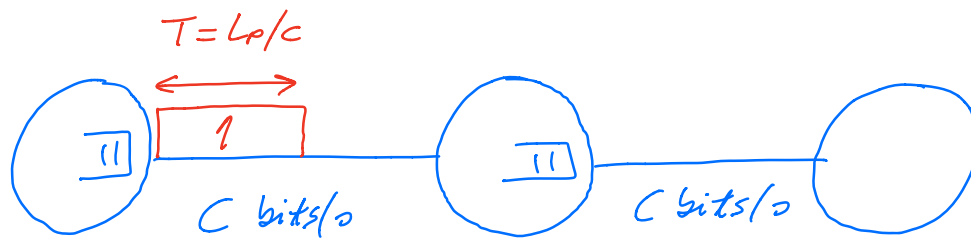
Delay in two cascaded point-to-point links

Assuming:

- ❖ Poisson arrivals
- ❖ Fixed size packets
- ❖ Independence between arrivals and lengths
- ❖ Infinite capacity buffers



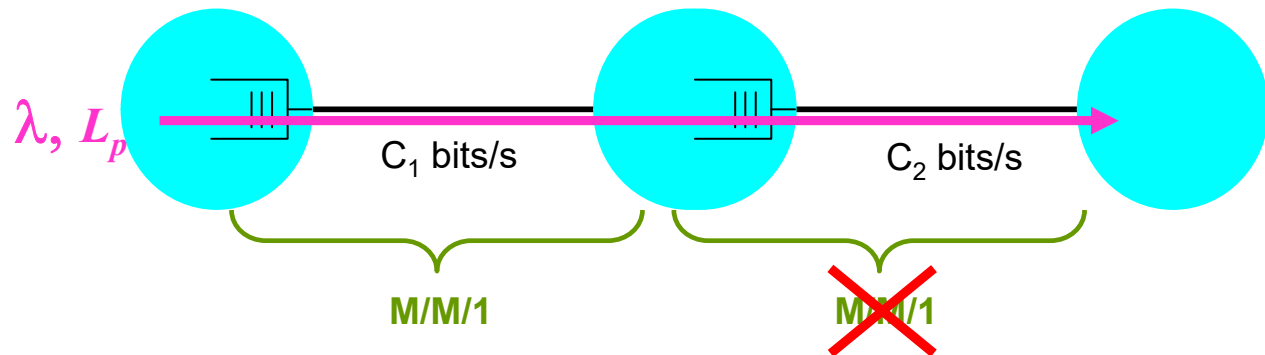
True average delay is $W = \frac{\rho}{2\mu(1-\rho)} + \frac{1}{\mu} + \frac{1}{\mu}$
end-to-end



Kleinrock approximation

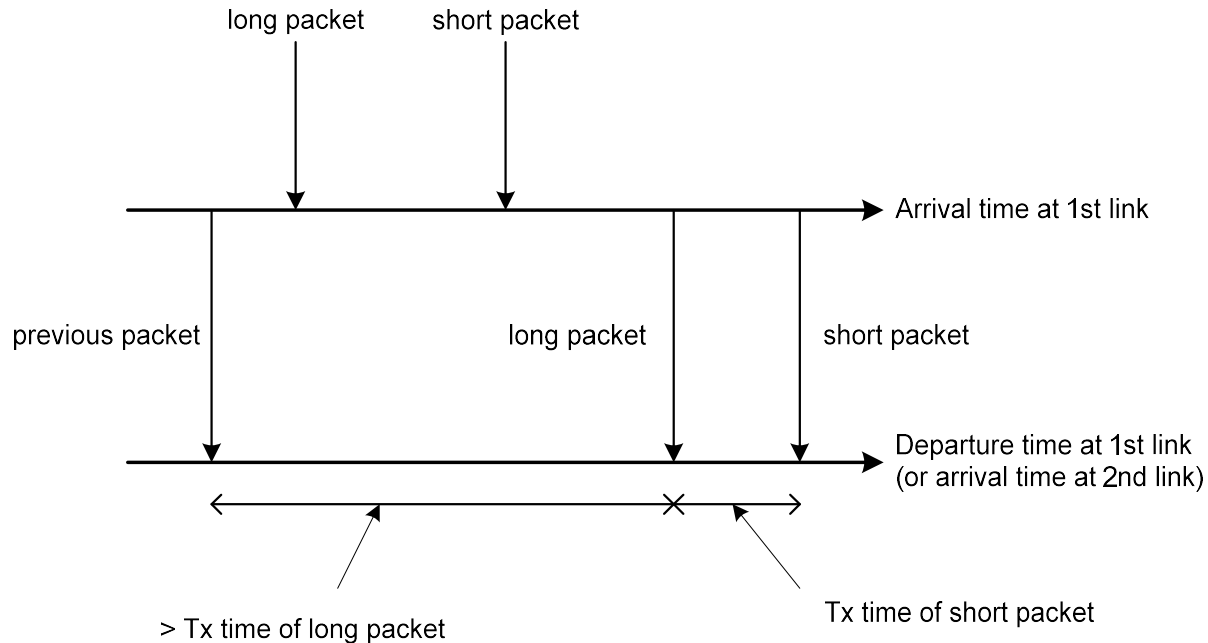
Assuming:

- ❖ Poisson arrivals
- ❖ Exponentially distributed packet sizes
- ❖ Independence between arrivals and lengths
- ❖ Infinite capacity buffers



- Kleinrock approximation: $W = \frac{1}{C_1/\bar{L}_p - \lambda} + \frac{1}{C_2/\bar{L}_p - \lambda}$
- Second and subsequent links are not M/M/1, but we assume they are – Kleinrock approximation

Kleinrock approximation

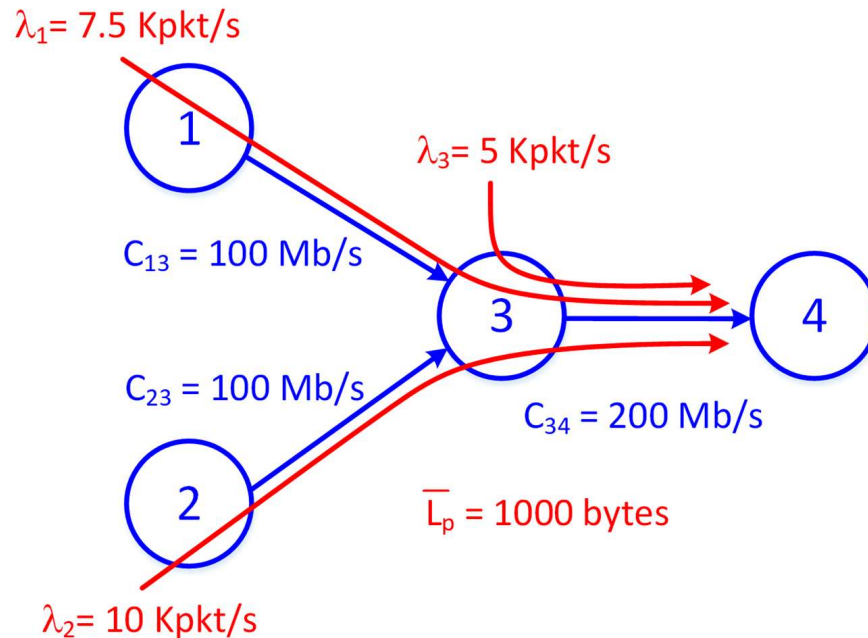


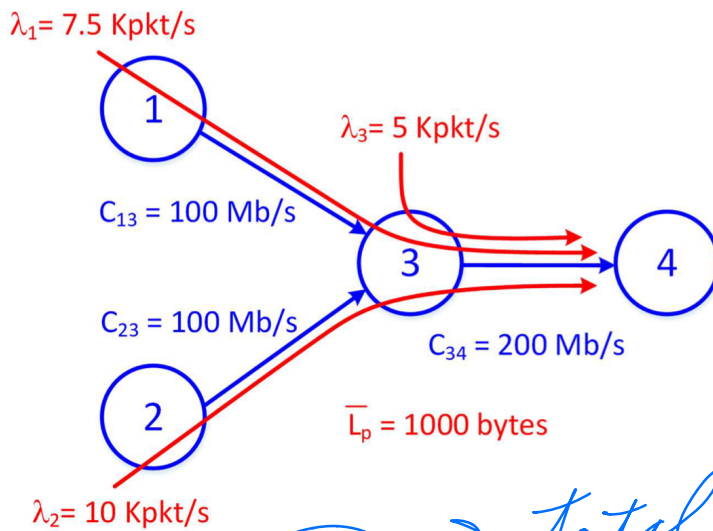
The inter-arrival time of two consecutive packets at the 2nd link is always $\geq$ Tx time of the 2nd packet in the 1st link

$\Rightarrow$ correlation between sizes and arrivals $\rightarrow$ violates M/M/1

Kleinrock approximation example

- Determine the average number of packets in the network, the average network delay and the average delay in each packet flow.





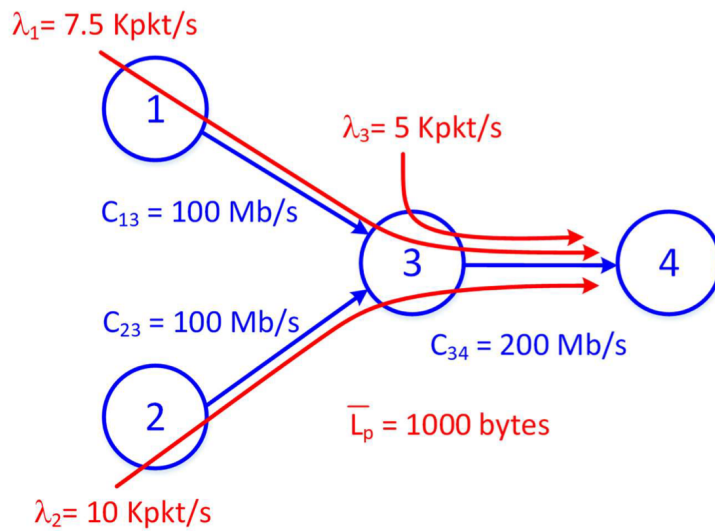
$$L_{ij} = \frac{\lambda_{ij}}{\mu_{ij} - \lambda_{ij}}$$

total rate offered to link (i,j)

$$\mu_{ij} = \frac{C_{ij}}{\bar{L}_p}$$

$$\mu_{13} = \mu_{23} = \frac{100 \text{ M}}{1 \text{ K} \times 8} = 12.5 \text{ Kpkt/s}$$

$$\mu_{34} = \frac{200 \text{ M}}{1 \text{ K} \times 8} = 25 \text{ Kpkt/s}$$

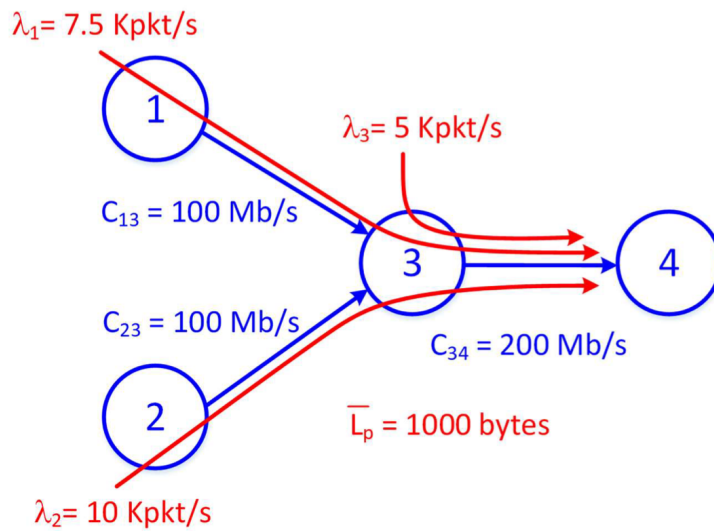


$$L = L_{13} + L_{23} + L_{34} = 14.5 \text{ pkt}$$

$$L_{13} = \frac{\lambda_{13}}{\mu_{13} - \lambda_{13}} = \frac{\lambda_1}{\mu_{13} - \lambda_1} = 9.5 \text{ pkt}$$

$$L_{23} = \frac{\lambda_{23}}{\mu_{23} - \lambda_{23}} = \frac{\lambda_2}{\mu_{23} - \lambda_2} = 4 \text{ pkt}$$

$$L_{34} = \frac{\lambda_{34}}{\mu_{34} - \lambda_{34}} = \frac{\lambda_1 + \lambda_2 + \lambda_3}{\mu_{34} - (\lambda_1 + \lambda_2 + \lambda_3)} = 9 \text{ pkt}$$



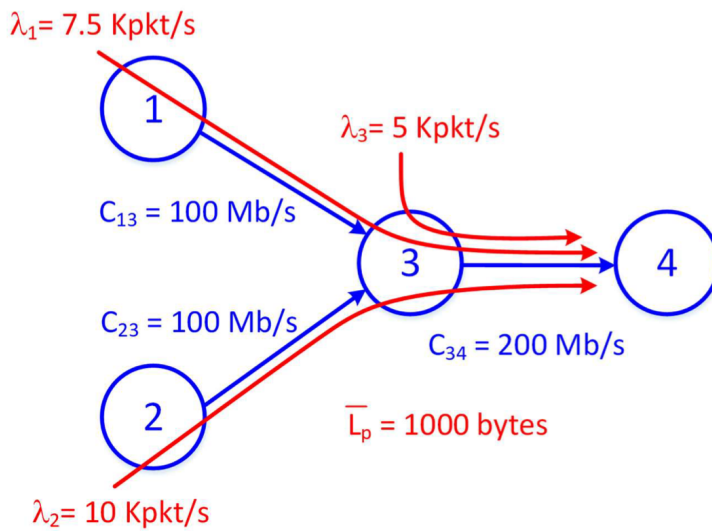
$$W = \frac{L}{\lambda_1 + \lambda_2 + \lambda_3} = 644. (4) \mu s$$

$$W_1 = W_{13} + W_{34}$$

$$W_2 = W_{23} + W_{34}$$

$$W_3 = W_{34}$$

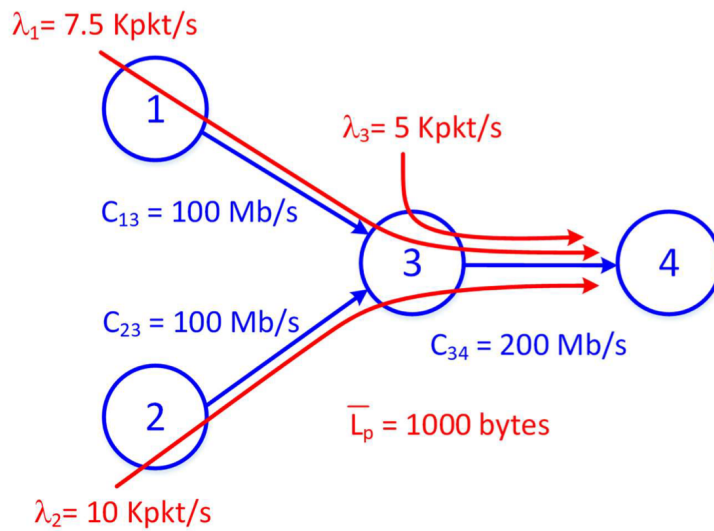
$$W_{ij} = \frac{1}{\mu_{ij} - \lambda_{ij}}$$



$$W_1 = \frac{1}{\mu_{13} - \lambda_1} + \frac{1}{\mu_{34} - (\lambda_1 + \lambda_2 + \lambda_3)} = 600 \mu s$$

$$W_2 = \frac{1}{\mu_{23} - \lambda_2} + \frac{1}{\mu_{34} - (\lambda_1 + \lambda_2 + \lambda_3)} = 800 \mu s$$

$$W_3 = \frac{1}{\mu_{34} - (\lambda_1 + \lambda_2 + \lambda_3)} = 400 \mu s$$

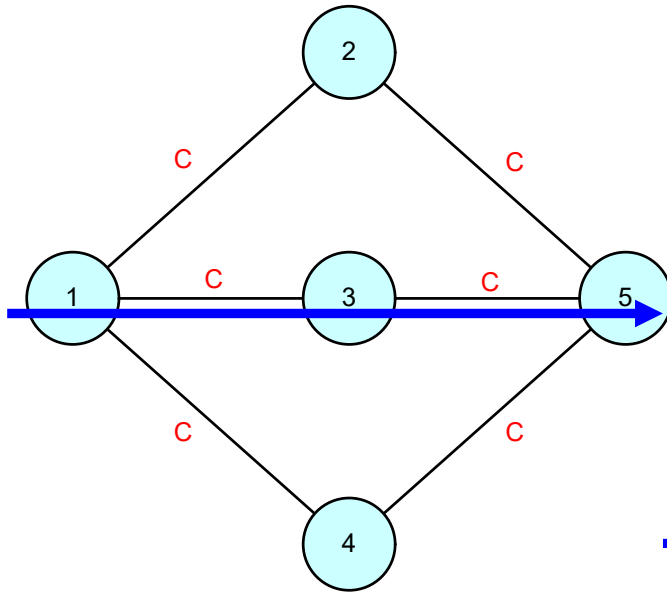


$$W_1, W_2, W_3 \rightarrow W$$

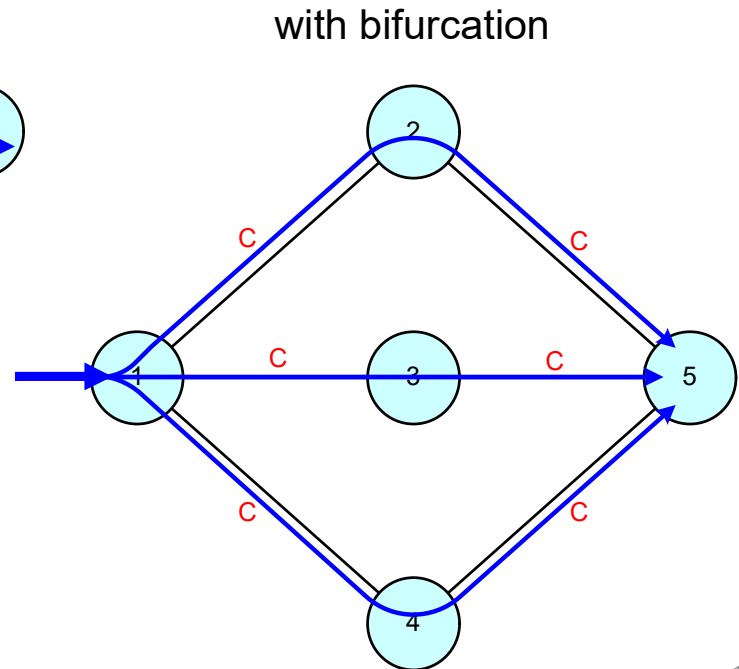
$$W = \frac{\lambda_1}{\sum_i \lambda_i} W_1 + \frac{\lambda_2}{\sum_i \lambda_i} W_2 + \frac{\lambda_3}{\sum_i \lambda_i} W_3 =$$

$$= 644. (4) \mu s$$

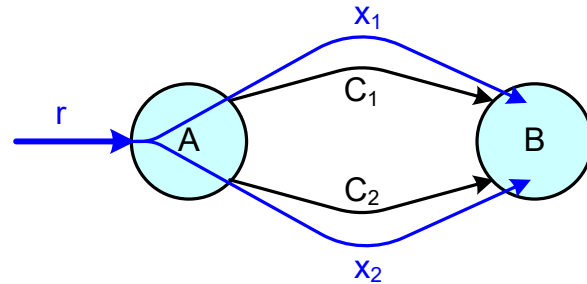
Traffic bifurcation



without bifurcation



Optimal routing – simple example



- Objective: minimize the mean packet delay

- M/M/1:
$$L = \frac{\lambda}{\mu - \lambda} = \frac{\lambda L_p}{\mu L_p - \lambda L_p} = \frac{x}{C - x}$$

λ – arrival rate in packets/s
 μ – service rate in packets/s
 L_p – average packet length
 x – arrival rate in bits/s
 C – link capacity in bits/s

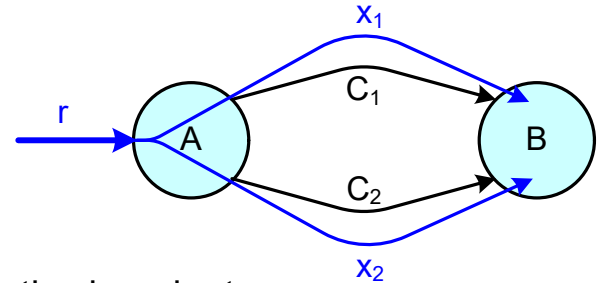
Note: minimization of W is equivalent to minimization of $L = rW$

Minimize:
$$L = \frac{x_1}{C_1 - x_1} + \frac{x_2}{C_2 - x_2}$$

restricted to: $r = x_1 + x_2$

$$x_1, x_2 \geq 0$$

Optimal routing – simple example



Replacing x_2 by $r - x_1$ and calculating the derivative in order to x_1 :

$$L = \frac{x_1}{C_1 - x_1} + \frac{r - x_1}{C_2 - (r - x_1)} \quad \frac{\partial L}{\partial x_1} = \frac{C_1}{(C_1 - x_1)^2} - \frac{C_2}{(C_2 - (r - x_1))^2}$$

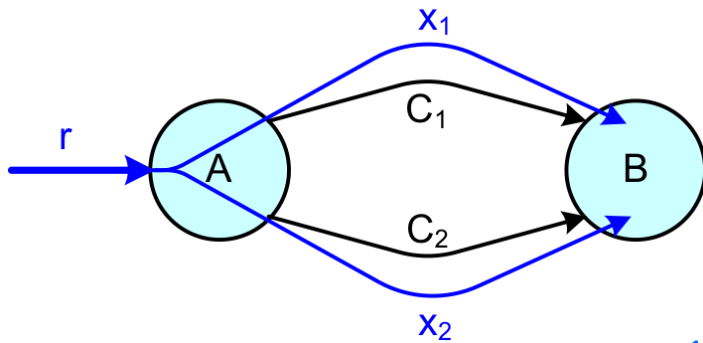
Making $\frac{\partial L}{\partial x_1} = \frac{C_1}{(C_1 - x_1)^2} - \frac{C_2}{(C_2 - (r - x_1))^2} = 0$

Results $x_1^* = \frac{\sqrt{C_1} \left[r - (C_2 - \sqrt{C_1 C_2}) \right]}{\sqrt{C_1} + \sqrt{C_2}} \quad x_2^* = \frac{\sqrt{C_2} \left[r - (C_1 - \sqrt{C_1 C_2}) \right]}{\sqrt{C_1} + \sqrt{C_2}}$

Assuming that $C_1 > C_2$ there are two cases:

$$r < C_1 - \sqrt{C_1 C_2} \rightarrow x_1^* = r, \quad x_2^* = 0$$

$$r > C_1 - \sqrt{C_1 C_2} \rightarrow 0 < x_1^* < r, \quad 0 < x_2^* < r$$



$$\begin{cases} L = \frac{x_1}{C_1 - x_1} + \frac{x_2}{C_2 - x_2} \\ r = x_1 + x_2 \end{cases}$$

$$L = \frac{x_1}{C_1 - x_1} + \frac{r - x_1}{C_2 - (r - x_1)}$$

$$\frac{\partial L}{\partial x_1} = \frac{1 \cdot (C_1 - x_1) - (-1) \cdot x_1}{(C_1 - x_1)^2} + \frac{(-1) \cdot (C_2 - r + x_1) - 1 \cdot (r - x_1)}{(C_2 - r + x_1)^2}$$

$$= \frac{C_1 - \cancel{x_1} + \cancel{x_1}}{(C_1 - x_1)^2} + \frac{-C_2 + \cancel{r} - \cancel{x_1} - \cancel{r} + \cancel{x_1}}{(C_2 - r + x_1)^2}$$

$$= \frac{C_1}{(C_1 - x_1)^2} - \frac{C_2}{(C_2 - r + x_1)^2} = 0$$

$$\frac{C_1}{(C_1 - x_1)^2} = \frac{C_2}{(C_2 - x + x_1)^2}$$

$$\frac{\sqrt{C_1}}{C_1 - x_1} = \frac{\sqrt{C_2}}{C_2 - x + x_1}$$

$$\sqrt{C_1} C_2 - \sqrt{C_1} x + \sqrt{C_1} x_1 = \sqrt{C_2} C_1 - \sqrt{C_2} x_1$$

$$(\sqrt{C_1} + \sqrt{C_2}) x_1 = \sqrt{C_2} C_1 - \sqrt{C_1} C_2 + \sqrt{C_1} x$$

$$x_1^* = \frac{\sqrt{C_1} (x - (C_2 - \sqrt{C_1 C_2}))}{\sqrt{C_1} + \sqrt{C_2}}$$

$$x_2^* = x - x_1^* = \frac{\sqrt{C_2} (x - (C_1 - \sqrt{C_1 C_2}))}{\sqrt{C_1} + \sqrt{C_2}}$$

$$x_1^* = \frac{\sqrt{C_1} \left[r - (C_2 - \sqrt{C_1 C_2}) \right]}{\sqrt{C_1} + \sqrt{C_2}}$$

$$x_2^* = \frac{\sqrt{C_2} \left[r - (C_1 - \sqrt{C_1 C_2}) \right]}{\sqrt{C_1} + \sqrt{C_2}}$$

$$C_2 - \sqrt{C_1 C_2} = C_2 \left(1 - \sqrt{\frac{C_1}{C_2}} \right) = C_2 \left(1 - \sqrt{\frac{C_1}{C_2}} \right)$$

$$C_1 > C_2 \Rightarrow \sqrt{\frac{C_1}{C_2}} > 1 \Rightarrow C_2 \left(1 - \sqrt{\frac{C_1}{C_2}} \right) < 0$$

$$\Rightarrow r - C_2 \left(1 - \sqrt{\frac{C_1}{C_2}} \right) > 0 \Rightarrow$$

$x_1^* > 0$ for any $r > 0$

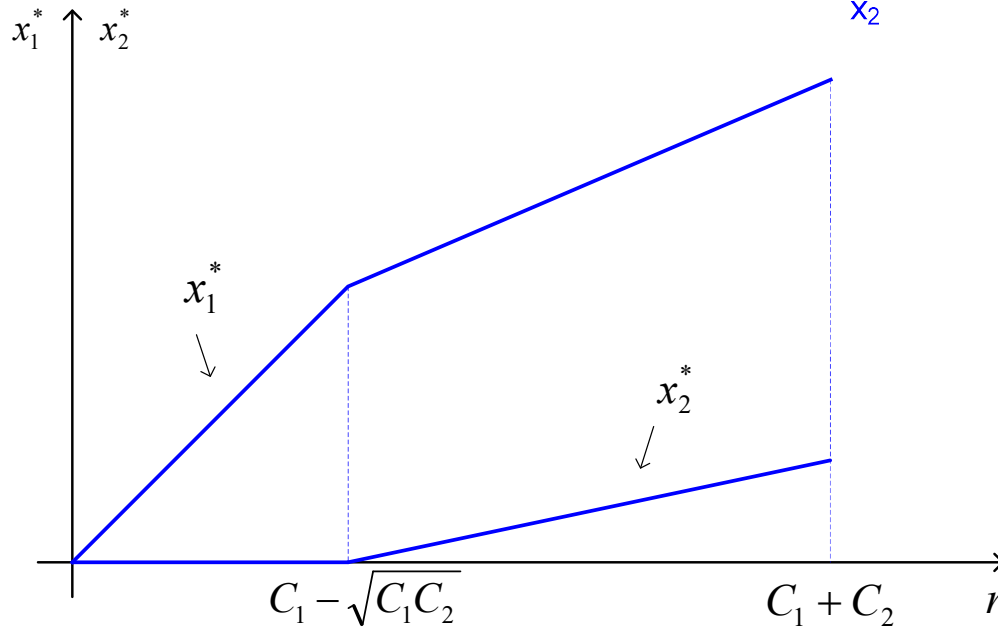
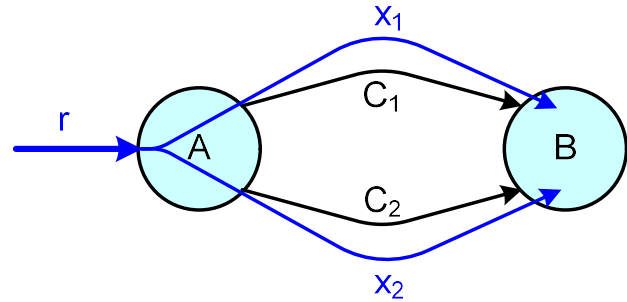
$$C_1 - \sqrt{C_1 C_2} = C_1 \left(1 - \sqrt{\frac{C_2}{C_1}} \right)$$

$$C_1 > C_2 \Rightarrow C_1 \left(1 - \sqrt{\frac{C_2}{C_1}} \right) > 0$$

$$\Rightarrow r - C_1 \left(1 - \sqrt{\frac{C_2}{C_1}} \right) \text{ can be } >, =, < \text{ than } 0$$

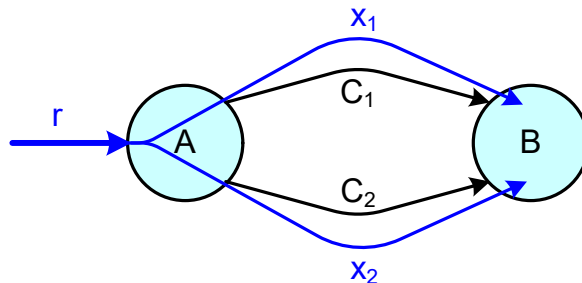
x_2^* can be $>, =, < \text{ than } 0$

Optimal routing – simple example



- For small r , optimal solution is to send all traffic through the higher capacity link

Optimal routing - simple example



$$dL = \frac{\partial L}{\partial x_1} dx_1 + \frac{\partial L}{\partial x_2} dx_2$$

$$x_1^*, x_2^* > 0$$

Remove Δ from x_1^*

Give Δ to x_2^*

$$dL = \frac{\partial L(\vec{x}^*)}{\partial x_1} (-\Delta) + \frac{\partial L(\vec{x}^*)}{\partial x_2} \Delta \geq 0 \quad \frac{\partial L(\vec{x}^*)}{\partial x_2} \geq \frac{\partial L(\vec{x}^*)}{\partial x_1}$$

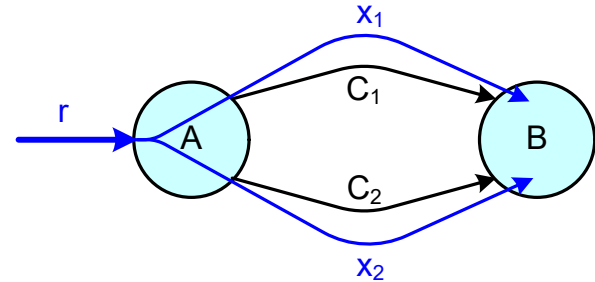
Remove Δ from x_2^*

Give Δ to x_1^*

$$dL = \frac{\partial L(\vec{x}^*)}{\partial x_1} \Delta + \frac{\partial L(\vec{x}^*)}{\partial x_2} (-\Delta) \geq 0 \quad \frac{\partial L(\vec{x}^*)}{\partial x_1} \geq \frac{\partial L(\vec{x}^*)}{\partial x_2}$$

$$\frac{\partial L(\vec{x}^*)}{\partial x_1} = \frac{\partial L(\vec{x}^*)}{\partial x_2}$$

Optimal routing - simple example



$$dL = \frac{\partial L(\bar{x}^*)}{\partial x_1} dx_1 + \frac{\partial L(\bar{x}^*)}{\partial x_2} dx_2$$

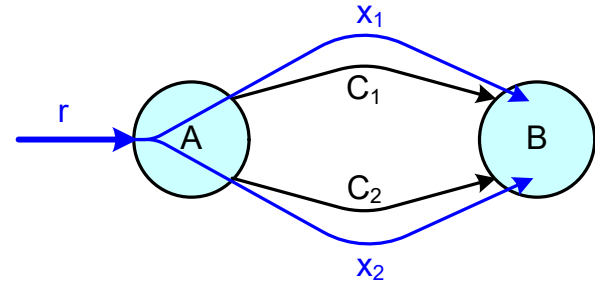
$$x_1^* > 0, x_2^* = 0$$

Remove Δ from x_1^*

Give Δ to x_2^*

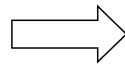
$$dL = \frac{\partial L(\bar{x}^*)}{\partial x_1} (-\Delta) + \frac{\partial L(\bar{x}^*)}{\partial x_2} \Delta \geq 0 \quad \frac{\partial L(\bar{x}^*)}{\partial x_2} \geq \frac{\partial L(\bar{x}^*)}{\partial x_1}$$

Optimal routing - simple example



$$L = \frac{x_1}{C_1 - x_1} + \frac{x_2}{C_2 - x_2} \quad \frac{\partial L}{\partial x_1} = \frac{C_1}{(C_1 - x_1)^2} \quad \frac{\partial L}{\partial x_2} = \frac{C_2}{(C_2 - x_2)^2}$$

$$\begin{aligned} C_1 &= 64 \text{ Kb/s} \\ C_2 &= 32 \text{ Kb/s} \end{aligned}$$



$$r_{\text{lim}} = C_1 - \sqrt{C_1 C_2} = 18.7 \text{ Kb/s}$$

$$r = 30 \quad x_1^* = 25.3 \quad x_2^* = 4.7$$

$$\frac{\partial L(\vec{x}^*)}{\partial x_1} = 0.043$$

$$\frac{\partial L(\vec{x}^*)}{\partial x_2} = 0.043$$

$$r = 10 \quad x_1^* = 10 \quad x_2^* = 0$$

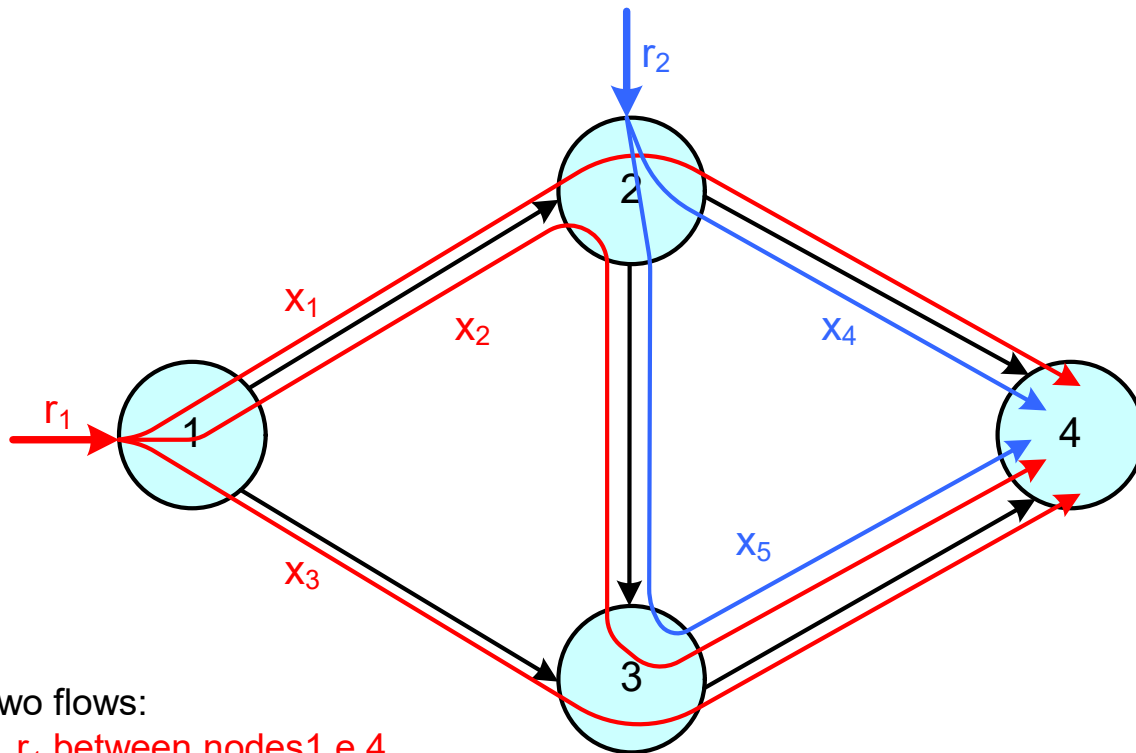
$$\frac{\partial L(\vec{x}^*)}{\partial x_1} = 0.022$$

$$\frac{\partial L(\vec{x}^*)}{\partial x_2} = 0.031$$

Optimal routing – general case

- Optimal solution is such that an optimal path flow is positive only on paths with minimum first derivative length (MFDL).
- There are two algorithms that obtain the optimal routing solution for more complex networks (where hand calculations are not possible)
 - Flow deviation algorithm (Frank-Wolfe)
 - Gradient projection algorithm (Bertsekas)
- These algorithm are based on shortest paths and determine a MFDL path, for each OD pair, at each iteration. In the projection algorithm, for each OD pair, at each iteration, an amount of flow is deviated from non-shortest paths to the MFDL path.

Optimal routing – general case

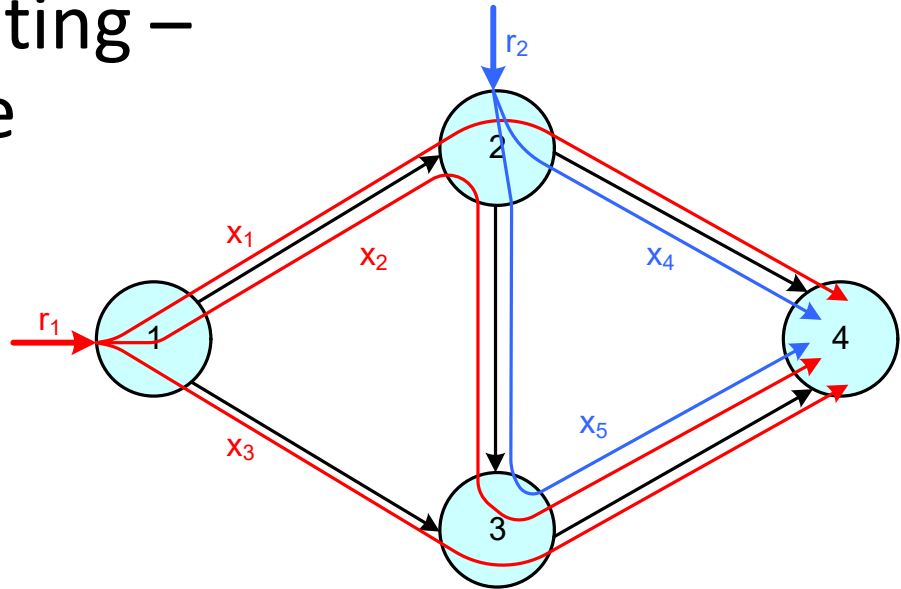


Two flows:

r_1 between nodes 1 e 4

r_2 between nodes 2 e 4

Optimal routing – general case



Minimize

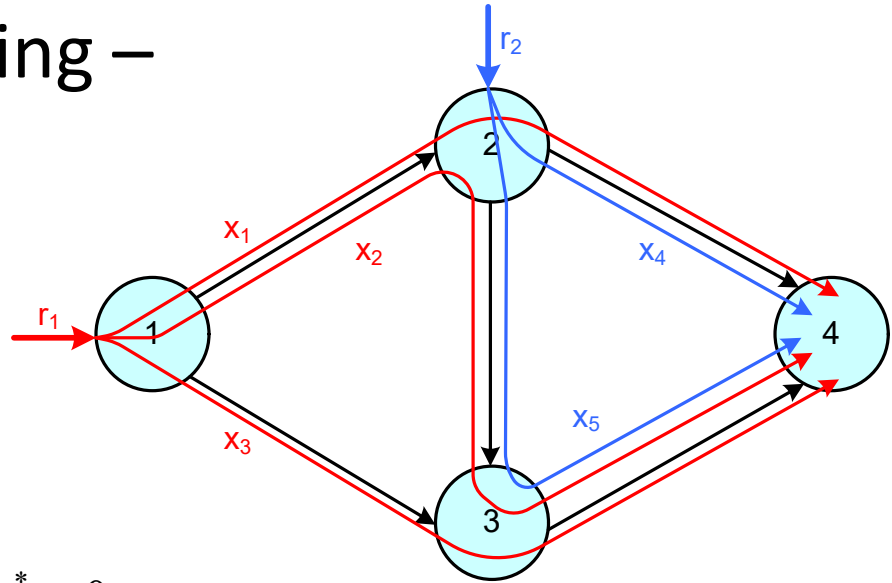
$$L = \frac{x_1 + x_2}{C_{12} - (x_1 + x_2)} + \frac{x_3}{C_{13} - x_3} + \frac{x_2 + x_5}{C_{23} - (x_2 + x_5)} + \frac{x_1 + x_4}{C_{24} - (x_1 + x_4)} + \frac{x_2 + x_3 + x_5}{C_{34} - (x_2 + x_3 + x_5)}$$

restricted to $r_1 = x_1 + x_2 + x_3$

$$r_2 = x_4 + x_5$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

Optimal routing – general case



solution with $x_1^*, x_2^*, x_3^*, x_4^*, x_5^* > 0$

$$\frac{\partial L(\bar{x}^*)}{\partial x_1} = \frac{\partial L(\bar{x}^*)}{\partial x_2} = \frac{\partial L(\bar{x}^*)}{\partial x_3}$$

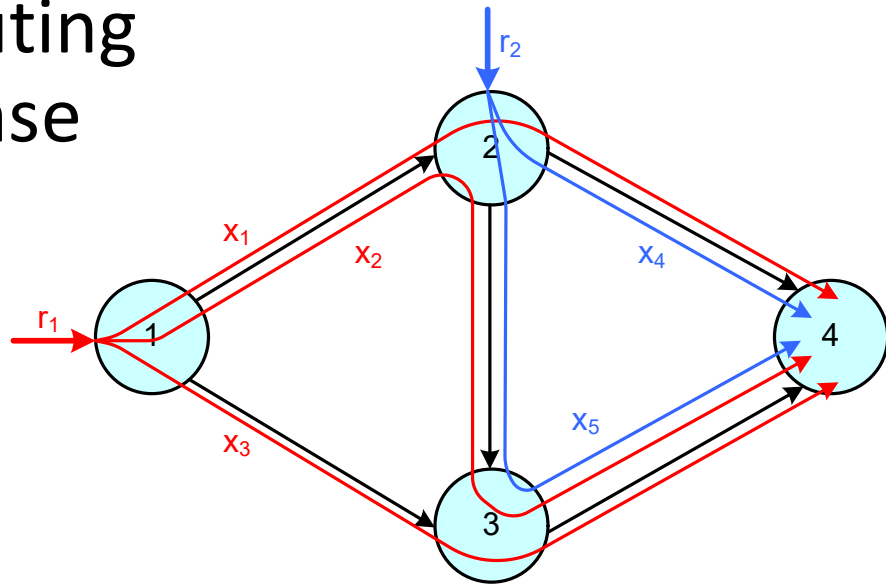
$$\frac{\partial L(\bar{x}^*)}{\partial x_4} = \frac{\partial L(\bar{x}^*)}{\partial x_5}$$

solution with $x_1^*, x_2^*, x_4^*, x_5^* > 0; x_3^* = 0$

$$\frac{\partial L(\bar{x}^*)}{\partial x_3} > \frac{\partial L(\bar{x}^*)}{\partial x_1} = \frac{\partial L(\bar{x}^*)}{\partial x_2}$$

$$\frac{\partial L(\bar{x}^*)}{\partial x_4} = \frac{\partial L(\bar{x}^*)}{\partial x_5}$$

Optimal routing – general case



Other optimal solution:

$$x_1^* - \Delta \quad x_2^* + \Delta \quad x_4^* + \Delta \quad x_5^* - \Delta$$

Optimal solution is not unique!